B.Adhi Shakthi

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Skills

- **Programming Languages** : Python, Javascript, Java
- **Libraries and Frameworks** : Fast API, React, Machine Learning, SVM (Support Vector Machine), Natural Language Processing (NLP), Deep Learning, Python Libraries
- **Tools** : NumPy, PyTorch, PostgreSQL, Google Colab, Docker, MySQL, Generative AI, Git, Prompt Engineering, GitHub, Matplotlib, Pandas, Unsupervised Learning, Supervised Learning, REST API, scikit-learn, AWS

Projects

PilotMaster - Retrieval Intelligence, Analysis and Benchmarking Platform

https://pilot-master.vercel.app/

- Built **PilotMaster**, a modular **retrieval intelligence platform** from first principles using **FastAPI, React, PostgreSQL, FAISS, Docker, and modern LLM frameworks**, integrating **DocPilot***(grounded document QA)*, **TracePilot***(retrieval observability and replayable execution tracing)*, and **GaugePilot***(retrieval benchmarking, ranking, experimentation and AI analysis)*.
  GitHub:https://github.com/AdhiShak3008/PilotMaster
- Engineered an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline supporting **multi-format document ingestion, OCR, semantic chunking, dense vector retrieval, BM25, Reciprocal Rank Fusion (RRF), cross-encoder reranking**, and **retrieval quality evaluation**.
- Developed **Production Mode** for grounded document intelligence and conversational document QA, alongside **Experimental Mode** for **retrieval strategy benchmarking, query rewriting, multi-query retrieval, parent-child retrieval, comparative model evaluation**, and experimentation.
- Implemented **retrieval observability** with **execution traces, chunk ranking analysis, retrieval lineage, grounding metrics, latency analytics, replay debugging**, and **hallucination risk analysis** for transparent AI system debugging.
- Built **GaugePilot**, an integrated **benchmarking workspace within the experimental mode** for evaluating retrieval configurations across multiple models using **leaderboards, interactive visualizations, automated insights**, and **optimization recommendations**.

Heart Disease RAG Assistant

https://github.com/AdhiShak3008/Heart-disease-RAG-APP

- Built an end-to-end AI-powered heart disease assistant using **PyTorch, FastAPI, React, and FAISS**, capable of classifying heart murmurs from multi-location phonocardiogram (PCG) recordings.
- Developed a deep learning inference pipeline with **RosaNet CNN**, spectrogram-based preprocessing, patient-level data splitting, weighted training, and REST APIs for real-time prediction.
- Implemented a **Retrieval-Augmented Generation (RAG)** pipeline with medical document ingestion, semantic chunking, vector search, and evidence-grounded responses sourced from trusted clinical references.
- Designed an interactive healthcare dashboard that combines AI predictions, confidence scores, retrieved medical evidence, and conversational explanations to support clinical decision-making.

AI Engineering Workflow Executor

https://github.com/AdhiShak3008/AI-Engineering-Workflow-Executor

- Built a deterministic multi-agent AI orchestration framework using FastAPI, Groq, LangGraph and Pydantic with Planner, Researcher, Summarizer, Validator, and Finalizer agents.

IT Support LLM using QLoRA Fine-Tuning

https://github.com/AdhiShak3008/IT-support-LLM-QLORA

- Fine-tuned Mistral-7B-Instruct using QLoRA, PEFT, TRL, and bitsandbytes to build a domain-specific IT support assistant.

- Implemented a complete fine-tuning pipeline including dataset preprocessing, 4-bit quantized LoRA training, benchmark evaluation, adapter-based inference, and Gradio deployment for real-time troubleshooting generation.

Credit Risk Analysis System
https://github.com/AdhiShak3008/Credit-Risk-Analysis-System
- Built a credit risk prediction system using Logistic Regression with reusable preprocessing, feature engineering, and versioned inference pipelines.
- Evaluated calibration, feature importance, and Top-X risk ranking to improve customer default prediction.

Credit Card Fraud Detection Decision System
https://github.com/AdhiShak3008/Credit-Card-Fraud-Detection
- Built a cost-aware fraud detection system using Logistic Regression, Random Forest, and XGBoost, achieving a PR-AUC of 0.94.
- Designed a decision engine for transaction blocking and OTP verification using threshold-based business cost optimization.

Experience

OCR Engineer Sep 2023 - Aug 2024
First American India
- Developed regex-driven extraction and validation pipelines for noisy OCR-generated American real estate documents using in-house OCR tool called DT-OCR and using PyTesseract as the fallback option.
- Improved document normalization, field-level validation, regex cleanups at document and global level and rule-based parsing for reliable downstream processing.

Intern Feb 2022 - Mar 2022
Defence Research and Development Laboratory
- Developed a two-dimensional finite-volume grid preprocessing utility in C for CFD solver-compatible structured grid conversion.

Achievements

Research Publication : PilotMaster: A Modular Retrieval Intelligence Platform for Retrieval Engineering, Observability and Benchmarking (https://www.researchgate.net/publication/408305845_PilotMaster_A_Modular_Retrieval_Intelligence_Platform_for_Retrieval_Engineering_Observability_and_Benchmarking)

Education

Scaler Academy x Woolf College - Neovarsity 2026
MS

CDAC Hyderabad 2023
Specialisation in Software Development

MVSR Engineering College 2022
BE/B.Tech/Bs